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JOULE-THOMSON CRYOCOOLER

Abstract

A Joule-Thomson cryocooler includes a pre-cooling cryocooler that includes a pre-cooling stage, a refrigerant circuit that includes a heat exchanger and a refrigerant pipe extending from the heat exchanger and cooled by the pre-cooling stage, and a heat conduction path that is provided separately from the refrigerant pipe and connects the pre-cooling stage to the heat exchanger to enable conductive cooling of the heat exchanger through the pre-cooling stage.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a bypass continuation of International PCT Application No. PCT/JP2023/038529, filed on Oct. 25, 2023, which claims priority to Japanese Patent Application No. 2022-187736, filed on Nov. 24, 2022, which are incorporated by reference herein in their entirety.

BACKGROUND

Technical Field

[0002] A certain embodiment of the present invention relates to a Joule-Thomson (JT) cryocooler.

Description of Related Art

[0003] In the related art, there has been known a JT cryocooler that includes a JT valve that enables cooling of a refrigerant gas using JT expansion, and a pre-cooling cryocooler such as a Gifford-McMahon (GM) cryocooler that pre-cools the refrigerant gas supplied to the JT valve. In order to cool the refrigerant gas through the JT expansion in the JT valve, the refrigerant gas needs to be pre-cooled to a temperature equal to or lower than an inversion temperature at which a JT coefficient becomes zero.

SUMMARY

[0004] According to an embodiment of the present invention, there is provided a Joule-Thomson cryocooler including a pre-cooling cryocooler that includes a pre-cooling stage, a refrigerant circuit that includes a heat exchanger and a refrigerant pipe extending from the heat exchanger and cooled by the pre-cooling stage, and a heat conduction path that is provided separately from the refrigerant pipe and connects the pre-cooling stage to the heat exchanger to enable conductive cooling of the heat exchanger through the pre-cooling stage.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a diagram schematically showing a cryogenic cooling device according to an embodiment.

[0006] FIG. 2 is a diagram schematically showing an exemplary configuration of a heat conduction path according to the embodiment.

[0007] FIG. 3 is a diagram schematically showing a modification example of the heat conduction path according to the embodiment.

DETAILED DESCRIPTION

[0008] When a JT cryocooler is started, so-called cool-down is performed in which the JT cryocooler is cooled from an environmental temperature (for example, a room temperature of about 300 K) to a target cryogenic temperature. A refrigerant gas supplied to a JT valve is cooled from the environmental temperature to a temperature equal to or lower than an inversion temperature by the cool-down. Since the cool-down is merely a preparation for the JT cryocooler to cool a desired object to be cooled to a cryogenic temperature, it is desirable that a time required for the cool-down is as short as possible.

[0009] It is desirable to shorten a cool-down time of the JT cryocooler.

[0010] Hereinafter, an embodiment for carrying out the present invention will be described in detail with reference to the drawings. In the description and the drawings, the same or equivalent components, members, and processes are denoted by the same reference numerals, and overlapping

description is omitted as appropriate. The scale and the shape of each of parts illustrated in the drawings are set for convenience to make the description easy to understand, and are not to be interpreted as limiting unless stated otherwise. The embodiment is merely an example and does not limit the scope of the present invention. All features described in the embodiment or combinations thereof are not necessarily essential to the present invention.

[0011] FIG. 1 is a diagram schematically showing a cryogenic cooling device **10** according to an embodiment. The cryogenic cooling device **10** includes a vacuum chamber **12**, a radiation shield **14**, and a JT cryocooler **18** for cooling an object **16** to be cooled.

[0012] The vacuum chamber **12** may be, for example, a cryostat, and is configured to provide a cryogenic vacuum environment therein. The vacuum chamber **12** is formed of a metal material such as stainless steel or other suitable high-strength material to withstand ambient pressure (for example, atmospheric pressure). The radiation shield **14**, a low-temperature section of the JT cryocooler **18**, and the object **16** to be cooled are disposed in the vacuum chamber **12**.

[0013] The radiation shield **14** is disposed in the vacuum chamber **12** to surround the low-temperature section of the JT cryocooler **18** and the object **16** to be cooled, and suppresses inflow of radiant heat from the vacuum chamber **12** to the JT cryocooler **18** and the object **16** to be cooled. The radiation shield **14** is formed of a high thermal conductivity metal material such as copper (for example, pure copper). An insulator such as a multi-layer insulator may be disposed between the vacuum chamber **12** and the radiation shield **14**.

[0014] The object to be cooled **16** may be, for example, a superconducting device such as a superconducting coil, a measuring device that operates better at a cryogenic temperature, or other devices for use at a cryogenic temperature. Alternatively, the object **16** to be cooled may be, for example, a cryogenic fluid such as liquid helium, and the JT cryocooler **18** may be used for recondensing a vaporized cryogenic fluid.

[0015] The JT cryocooler **18** includes a pre-cooling cryocooler **20** and a refrigerant circuit **40** including a JT valve **30** and a final heat exchanger **32**. A refrigerant flowing through the refrigerant circuit **40** is pre-cooled by the pre-cooling cryocooler **20**, further cooled by JT expansion in the JT valve **30**, and is supplied to the final heat exchanger **32**. The object **16** to be cooled is cooled by heat exchange with the final heat exchanger **32**. The cooled refrigerant is collected from the final heat exchanger **32**, pressurized by a compressor described below, pre-cooled by the pre-cooling cryocooler **20** again, and is supplied to the JT valve **30**. In this way, the refrigerant circulates through the refrigerant circuit **40**. The JT cryocooler **18** is capable of cooling the final heat exchanger **32** to a temperature range of, for example, about 4 K or lower (for example, 1 K to 4 K), and thus, the object **16** to be cooled can be cooled to the temperature range.

[0016] The pre-cooling cryocooler **20** is, for example, a two-stage GM cryocooler. The pre-cooling cryocooler **20** includes a first compressor **21** and an expander **22** also referred to as a cold head. The expander **22** includes a drive unit **23**, a first cylinder **24**, a first pre-cooling stage **25**, a second cylinder **26**, and a second pre-cooling stage **27**. The first compressor **21** is disposed in an ambient environment (for example, room temperature and atmospheric pressure environment), that is, outside the vacuum chamber **12**. The expander **22** is installed in the vacuum chamber **12** such that the drive unit **23** is disposed outside the vacuum chamber **12** and the cylinders and the pre-cooling stages are disposed inside the vacuum chamber **12**.

[0017] The first cylinder **24** connects the first pre-cooling stage **25** to the drive unit **23**, so that the first pre-cooling stage **25** is structurally supported by the drive unit **23**. The second cylinder **26** connects the second pre-cooling stage **27** to the first pre-cooling stage **25**, so that the second pre-cooling stage **27** is structurally supported by the first pre-cooling stage **25**. The first cylinder **24** and the second cylinder **26** extend coaxially, and the drive unit **23**, the first cylinder **24**, the first pre-cooling stage **25**, the second cylinder **26**, and the second pre-cooling stage **27** are linearly arranged in a line in this order. Typically, the first pre-cooling stage **25** and the second pre-cooling stage **27** are formed of a high thermal conductivity metal material such as copper (for example, pure

copper), and the first cylinder **24** and the second cylinder **26** are formed of other metal materials such as stainless steel.

[0018] A first displacer and a second displacer (not shown) are disposed reciprocally in an inside of each of the first cylinder **24** and the second cylinder **26**. A first regenerator and a second regenerator (not shown) are incorporated into the first displacer and the second displacer, respectively. In addition, the drive unit **23** includes a drive mechanism (not shown) such as a motor for reciprocating the first displacer and the second displacer. The drive mechanism includes a flow path switching mechanism that switches a flow path of a refrigerant gas to periodically repeat supply and exhaust of the refrigerant gas to and from an inside of the expander **22**. A refrigerant gas of the pre-cooling cryocooler **20** is usually a helium gas, but other appropriate gases may be used.

[0019] The first compressor **21** is configured to collect the refrigerant gas from the expander **22**, to pressurize the collected refrigerant gas, and to supply the refrigerant gas to the expander **22** again. By circulating the refrigerant gas between the first compressor **21** and the expander **22** with an appropriate combination of a pressure fluctuation and a volume fluctuation of the refrigerant gas in the expander **22**, a thermodynamic cycle (for example, a GM cycle) for cold generation is configured, and the expander **22** can provide cryogenic cooling.

[0020] The first pre-cooling stage **25** is cooled to a first cooling temperature, and the second pre-cooling stage **27** is cooled to a second cooling temperature lower than the first cooling temperature. The first cooling temperature may be selected from a temperature range of, for example, 50 K or higher and 150 K or lower. The second cooling temperature may be selected from a temperature range of, for example, 10 K or higher and 25 K or lower.

[0021] The radiation shield **14** is thermally coupled to the first pre-cooling stage **25** by being in physical contact with the first pre-cooling stage **25**, or is thermally coupled to the first pre-cooling stage **25** via a heat transfer member. Therefore, the radiation shield **14** is cooled to the first cooling temperature by the first pre-cooling stage **25**.

[0022] The refrigerant circuit **40** includes a second compressor **41**, a heat exchanger group **42**, and a refrigerant supply line **44** and a refrigerant collection line **46** that connect these components in addition to the JT valve **30** and the final heat exchanger **32**. A refrigerant gas that circulates through the refrigerant circuit **40** is usually a helium gas, but other appropriate gases may be used. The refrigerant circuit **40** is not limited to the specific configuration described herein and can adopt various typical configurations as appropriate.

[0023] The second compressor **41** is configured to pressurize the refrigerant gas collected from the refrigerant collection line **46** and to send the collected refrigerant gas to the refrigerant supply line **44**. For understanding, a direction in which the refrigerant flows is indicated by an arrow in FIG. **1**. The second compressor **41** serves as a refrigerant source that circulates the refrigerant in the refrigerant circuit **40**. The second compressor **41** is disposed outside the vacuum chamber **12**.

[0024] The heat exchanger group **42** in the refrigerant circuit **40** is disposed between the second compressor **41** and the final heat exchanger **32**. The heat exchanger group **42** is composed of a series of counterflow heat exchangers (**42a** to **42c**), and has a three-stage configuration of a first heat exchanger **42a**, a second heat exchanger **42b**, and a third heat exchanger **42c** in the present embodiment. The first heat exchanger **42a** is disposed between the vacuum chamber **12** and the radiation shield **14**, that is, in a space inside the vacuum chamber **12** and outside the radiation shield **14**. The second heat exchanger **42b**, the third heat exchanger **42c**, and the final heat exchanger **32** are disposed inside the radiation shield **14**.

[0025] The first heat exchanger **42a** cools a high-temperature (for example, a room temperature, for example, approximately 300 K) refrigerant gas flowing into the vacuum chamber **12** from the outside of the vacuum chamber **12**. The second heat exchanger **42b** further cools the refrigerant cooled by the first heat exchanger **42a** and the first pre-cooling stage **25**. The third heat exchanger **42c** further cools the refrigerant cooled by the second heat exchanger **42b** and the second pre-cooling stage **27**.

[0026] The refrigerant supply line **44** connects a discharge side of the second compressor **41** to a refrigerant inlet of the final heat exchanger **32**, and the refrigerant collection line **46** connects a refrigerant outlet of the final heat exchanger **32** to a suction side of the second compressor **41**. The refrigerant supply line **44** includes a high pressure side flow path of each of the first heat exchanger **42a**, the second heat exchanger **42b**, and the third heat exchanger **42c**, and the refrigerant collection line **46** includes a low pressure side flow path of each of the first heat exchanger **42a**, the second heat exchanger **42b**, and the third heat exchanger **42c**. The refrigerant flowing through the high pressure side flow path can be cooled by heat exchange between the high pressure side flow path and the low pressure side flow path in each heat exchanger. The high pressure side flow path and the low pressure side flow path may be called a high temperature side flow path and a low temperature side flow path, respectively.

[0027] In addition, the refrigerant supply line **44** includes a first refrigerant pipe **44a** and a second refrigerant pipe **44b**. The refrigerant pipes are formed of, for example, a high thermal conductivity metal material such as copper (for example, pure copper).

[0028] The first refrigerant pipe **44a** extends from the first heat exchanger **42a** via the first pre-cooling stage **25** to the second heat exchanger **42b**. The first refrigerant pipe **44a** connects the high pressure side flow path of the first heat exchanger **42a** to the high pressure side flow path of the second heat exchanger **42b**. The first refrigerant pipe **44a** is thermally coupled to the first pre-cooling stage **25**, and the refrigerant flowing through the first refrigerant pipe **44a** is cooled by the first pre-cooling stage **25**. The first refrigerant pipe **44a** may be fixed to the first pre-cooling stage **25** in a state of being wound around an outer peripheral surface of the first pre-cooling stage **25**.

[0029] The second refrigerant pipe **44b** extends from the second heat exchanger **42b** via the second pre-cooling stage **27** to the third heat exchanger **42c**. The second refrigerant pipe **44b** connects the high pressure side flow path of the second heat exchanger **42b** to the high pressure side flow path of the third heat exchanger **42c**. The second refrigerant pipe **44b** is thermally coupled to the second pre-cooling stage **27**, and the refrigerant flowing through the second refrigerant pipe **44b** is cooled by the second pre-cooling stage **27**. The second refrigerant pipe **44b** may be fixed to the second pre-cooling stage **27** in a state of being wound around an outer peripheral surface of the second pre-cooling stage **27**.

[0030] The JT valve **30** is disposed between the last heat exchanger of the heat exchanger group **42** (in the present example, the third heat exchanger **42c**) and the final heat exchanger **32** in the refrigerant supply line **44**. The high pressure side flow path of the third heat exchanger **42c** is connected to the refrigerant inlet of the final heat exchanger **32** via the JT valve **30**. The JT valve **30** is a fixed orifice in the present embodiment. However, the JT valve **30** may be a variable orifice of which an opening degree is adjustable.

[0031] In the steady operation of the JT cryocooler **18**, the refrigerant flows through the refrigerant circuit **40** as follows. The high-pressure refrigerant compressed by the second compressor **41** is first supplied to the high pressure side flow path of the first heat exchanger **42a**. The high-pressure refrigerant flowing through the high pressure side flow path of the first heat exchanger **42a** is cooled by heat exchange with the returning low-pressure refrigerant flowing through the low pressure side flow path of the first heat exchanger **42a**. The high-pressure refrigerant cooled by the first heat exchanger **42a** flows into the first refrigerant pipe **44a**.

[0032] The high-pressure refrigerant flowing through the first refrigerant pipe **44a** is cooled by the first pre-cooling stage **25** of the pre-cooling cryocooler **20** and is sent into the high pressure side flow path of the second heat exchanger **42b**. The high-pressure refrigerant flowing through the high pressure side flow path of the second heat exchanger **42b** is cooled by heat exchange with the returning low-pressure refrigerant flowing through the low pressure side flow path of the second heat exchanger **42b**. The high-pressure refrigerant cooled by the second heat exchanger **42b** flows into the second refrigerant pipe **44b**.

[0033] The high-pressure refrigerant flowing through the second refrigerant pipe **44b** is cooled by

the second pre-cooling stage **27** of the pre-cooling cryocooler **20** and is sent into the high pressure side flow path of the third heat exchanger **42c**. The high-pressure refrigerant flowing through the high pressure side flow path of the third heat exchanger **42c** is cooled by heat exchange with the returning low-pressure refrigerant flowing through the low pressure side flow path of the third heat exchanger **42c**. In this way, the high-pressure refrigerant is cooled to a temperature (that is, a temperature equal to or lower than an inversion temperature) at which the JT effect is expected, and is sent to the JT valve **30**.

[0034] When the cooled high-pressure refrigerant passes through the JT valve **30**, the cooled high-pressure refrigerant becomes a low-pressure refrigerant in a mist-like gas-liquid mixed state due to the Joule-Thomson effect, and generates a cooling capacity in a temperature range of the liquefied refrigerant. The mist-like low-pressure refrigerant is sent to the final heat exchanger **32**. As described above, in a case where the refrigerant is helium, the final heat exchanger **32** can be cooled to a liquid helium temperature range. The final heat exchanger **32** can cool the object **16** to be cooled to the temperature through heat exchange with the object **16** to be cooled.

[0035] When cooling the final heat exchanger **32**, the mist-like low-pressure refrigerant evaporates and vaporizes again. The unliquefied refrigerant and the refrigerant that is vaporized by evaporation in the JT valve **30** are returned to the low pressure side flow path of the third heat exchanger **42c**. The low-pressure refrigerant flows through the refrigerant collection line **46** in the order of the third heat exchanger **42c**, the second heat exchanger **42b**, and the first heat exchanger **42a**. In this case, the low-pressure refrigerant is heated while cooling the high-pressure refrigerant in each of the heat exchangers (**42c**, **42b**, and **42a**) as described above. In this way, the low-pressure refrigerant that has returned to the room temperature exits the vacuum chamber **12**, is collected in the second compressor **41**, and is compressed again.

[0036] In this way, the cryogenic cooling device **10** can cool the object **16** to be cooled to a desired temperature lower than the second cooling temperature of the pre-cooling cryocooler **20**, for example, about 4 K or lower (for example, 1 K to 4 K).

[0037] Incidentally, when the JT cryocooler **18** is started, the JT cryocooler **18** is cooled from the environmental temperature (for example, a room temperature of about 300 K) to a target cryogenic temperature (for example, a minimum attainable temperature of less than 4 K). This initial cooling is also called cool-down. The refrigerant gas supplied to the JT valve **30** is cooled from the environmental temperature to a temperature equal to or lower than the inversion temperature by cool-down. Since the cool-down is merely a preparation for the JT cryocooler **18** to cool the object **16** to be cooled to a cryogenic temperature, it is desirable that a time required for the cool-down is as short as possible.

[0038] The JT valve **30** is designed to provide an optimal JT flow rate at a cryogenic temperature. Therefore, a flow rate of the refrigerant gas that can pass through the JT valve **30** in an initial stage of the cool-down where the refrigerant gas temperature is high can be considerably small. This can significantly increase a time required for the cool-down.

[0039] In a case where the JT valve **30** is the variable orifice, the JT valve **30** is opened wider than an optimal opening degree at the cryogenic temperature at the start of the cool-down to secure a large flow rate, and then the opening degree of the JT valve **30** is reduced as the cooling proceeds, thereby shortening the cool-down time. However, this measure is time-consuming. For example, a service technician may need to adjust the opening degree of the JT valve **30** to an appropriate level. In addition, in a case where the JT valve **30** is the fixed orifice, this measure cannot be adopted (as a result, it may take more than twice as long to cool the JT cryocooler **18** to a minimum attainable temperature (for example, approximately 4 K) compared to a case of the variable orifice).

[0040] In a cryocooler of an existing design, in general, a pre-cooling cryocooler and a heat exchanger of a JT refrigerant circuit are disposed so as not to be thermally connected to each other in order to prevent heat from entering the JT refrigerant circuit from the pre-cooling cryocooler during a normal operation. A pre-cooling stage of the pre-cooling cryocooler is supported by a

surrounding structure with an insulating material such as fiber reinforced plastic, and there is substantially no heat conduction path from the pre-cooling stage to the heat exchanger.

[0041] This embodiment aims to shorten the time required for the cool-down by using the pre-cooling cryocooler **20**. The JT cryocooler **18** includes a heat conduction path **48** that connects at least one pre-cooling stage and at least one heat exchanger of the pre-cooling cryocooler **20** to enable conductive cooling of at least one heat exchanger of the heat exchanger group **42** through the pre-cooling cryocooler **20**. The heat conduction path **48** may connect the second pre-cooling stage **27** to at least one of the second heat exchanger **42b** and the third heat exchanger **42c** to enable conductive cooling of at least one of the second heat exchanger **42b** and the third heat exchanger **42c** through the second pre-cooling stage **27**.

[0042] For example, the JT cryocooler **18** may include a first heat conduction path **48a** that connects the second pre-cooling stage **27** to the second heat exchanger **42b** to enable conductive cooling of the second heat exchanger **42b** through the second pre-cooling stage **27**. In addition, or instead of this, the JT cryocooler **18** may include a second heat conduction path **48b** that connects the second pre-cooling stage **27** to the third heat exchanger **42c** to enable conductive cooling of the third heat exchanger **42c** through the second pre-cooling stage **27**.

[0043] In order to prevent a heat load from being transferred from a high-temperature section to a low-temperature section of the JT cryocooler **18** when the cool-down is completed and the JT cryocooler **18** is in the normal operation, the heat conduction path **48** is installed such that a temperature difference between both ends of the heat conduction path **48** is as small as possible during the normal operation of the JT cryocooler **18**. The heat conduction path **48** may be installed such that the temperature difference between both ends of the heat conduction path **48** is, for example, less than 5 K or less than 3 K during the normal operation of the JT cryocooler **18**.

[0044] As an example, as shown in FIG. **1**, the first heat conduction path **48a** may be connected to a low temperature side of the second heat exchanger **42b**. In the exemplary JT cryocooler **18**, it is expected that the low temperature side of the second heat exchanger **42b** and the second pre-cooling stage **27** of the pre-cooling cryocooler **20** have similar temperatures (for example, about 12 K) during the normal operation of the JT cryocooler **18**.

[0045] On the other hand, a high temperature side of the second heat exchanger **42b** is assumed to have a temperature similar to that of the first pre-cooling stage **25** of the pre-cooling cryocooler **20**. Assuming that the first heat conduction path **48a** connects the high temperature side of the second heat exchanger **42b** and the second pre-cooling stage **27** of the pre-cooling cryocooler **20**, during the normal operation of the JT cryocooler **18**, a temperature difference corresponding to a temperature difference between the first pre-cooling stage **25** and the second pre-cooling stage **27** occurs between both ends of the first heat conduction path **48a**, which increases a heat load on the second pre-cooling stage **27** and may have an undesirable effect on the refrigeration performance of the JT cryocooler **18**.

[0046] For similar reasons, as shown in FIG. **1**, the second heat conduction path **48b** may be connected to a low temperature side of the third heat exchanger **42c**. Instead of this, or in addition to this, the second heat conduction path **48b** may be connected to a high temperature side of the third heat exchanger **42c**.

[0047] The refrigerant pipe (for example, the first refrigerant pipe **44a** or the second refrigerant pipe **44b**) that constitutes the refrigerant circuit **40** structurally connects the pre-cooling stage and the heat exchanger, but a cross-sectional area perpendicular to a pipe axial direction is considerably small, so that it is not possible to achieve sufficient heat conduction to enable reduction in the cool-down time.

[0048] Therefore, the heat conduction path **48** is provided separately from the refrigerant pipe that constitutes the refrigerant circuit **40**. That is, the first heat conduction path **48a** is provided separately from the second refrigerant pipe **44b**, and thermally couples the second pre-cooling stage **27** and the second heat exchanger **42b** to each other. The second heat conduction path **48b** is

provided separately from the second refrigerant pipe **44b**, and thermally couples the second pre-cooling stage **27** and the third heat exchanger **42c** to each other.

[0049] The heat conduction path **48** is formed of one or more heat conduction members. Such a heat conduction member is formed of, for example, a metal material having a high thermal conductivity, such as copper (for example, pure copper or a copper alloy) or aluminum (for example, pure aluminum or an aluminum alloy), or other high thermal conductivity material. The heat conduction member may be formed of, for example, a material having a higher thermal conductivity than stainless steel (for example, SUS304).

[0050] FIG. **2** is a diagram schematically showing an exemplary configuration of the heat conduction path according to the embodiment. As shown in the figure, the first heat conduction path **48a** includes a stage extension component **50** and a heat conduction plate **52** as an example of the heat conduction member.

[0051] The stage extension component **50** is a tubular (for example, cylindrical) heat conduction member formed of a high thermal conductivity material (for example, pure copper), and is thermally coupled to the second pre-cooling stage **27** at one end of the stage extension component **50** and is thermally coupled to the heat conduction plate **52** at the other end. For example, a flange formed at one end of the stage extension component **50** may be fixed to the second pre-cooling stage **27** by a fastening member such as a bolt, and a flange formed at the other end of the stage extension component **50** may be fixed to the heat conduction plate **52**.

[0052] The heat conduction plate **52** is a flat plate-shaped heat conduction member formed of a high thermal conductivity material (for example, an aluminum alloy), and the stage extension component **50** and a low temperature end of the second heat exchanger **42b** are thermally coupled by the heat conduction plate **52**. The stage extension component **50** and the second heat exchanger **42b** may be disposed on the same side of the heat conduction plate **52** and fixed to the same surface of the heat conduction plate **52**.

[0053] The second heat exchanger **42b** may have a typical configuration including a tubular casing. The second heat exchanger **42b** may include a mandrel coaxially disposed inside the casing, and a pipe wound around an outer peripheral surface of the mandrel and disposed in a tubular cavity between the mandrel and the casing. The pipe may be used as a high pressure side flow path of the second heat exchanger **42b**, and the tubular cavity may be used as the low pressure side flow path of the second heat exchanger **42b**.

[0054] As shown in the figure, the second refrigerant pipe **44b** and the refrigerant collection line **46** penetrate a low temperature side end plate of the casing of the second heat exchanger **42b**. As described with reference to FIG. **1**, the second refrigerant pipe **44b** is connected to the high pressure side flow path of the second heat exchanger **42b**, and the refrigerant collection line **46** is connected to the low pressure side flow path of the second heat exchanger **42b**. In addition, the second refrigerant pipe **44b** is attached to the second pre-cooling stage **27** by being wound around the second pre-cooling stage **27** to enable heat exchange with the second pre-cooling stage **27**.

[0055] Although not shown in FIG. **2**, the third heat exchanger **42c** may be thermally coupled to the second pre-cooling stage **27** via the heat conduction plate **52** and the stage extension component **50** in the same manner. The third heat exchanger **42c** may also have a tubular outer shape similarly to the second heat exchanger **42b**, and a low temperature end thereof may be fixed to the heat conduction plate **52**. As an exemplary disposition, the third heat exchanger **42c** may be disposed on a front side or a rear side of the second heat exchanger **42b** with respect to a paper surface of FIG. **2**, in parallel with the second heat exchanger **42b**.

[0056] As described above, according to the embodiment, the second heat exchanger **42b** and the third heat exchanger **42c** are connected to the second pre-cooling stage **27** of the pre-cooling cryocooler **20** by the heat conduction path **48**. During the cool-down, cooling of the heat exchanger group **42** of the JT cryocooler **18** can be facilitated by using the conductive cooling via the heat conduction path **48** in addition to the refrigerant gas circulating through the refrigerant circuit **40**.

Therefore, the cool-down time of the JT cryocooler **18** can be shortened.

[0057] According to the verification of the present inventor, the existing design in which the heat conduction path **48** is not provided requires approximately 24 hours to cool down, whereas in the embodiment in which the heat conduction path **48** is provided, it is possible to complete the cool-down in approximately 12 hours.

[0058] FIG. **3** is a diagram schematically showing a modification example of the heat conduction path according to the embodiment. As shown in the figure, the third heat exchanger **42c** may be thermally coupled to the second pre-cooling stage **27** by the second heat conduction path **48b**. Note that the second heat conduction path **48b** may include a thermal resistance element **60** in addition to the heat conduction member (for example, the stage extension component **50** and the heat conduction plate **52**). The second heat conduction path **48b** may connect the second pre-cooling stage **27** to the third heat exchanger **42c** via the thermal resistance element **60**.

[0059] The thermal resistance element **60** may be a spacer formed of a metal material or other material having a lower thermal conductivity than the heat conduction member, and the third heat exchanger **42c** may be fixed to the heat conduction plate **52** so as to sandwich the thermal resistance element **60** between the low temperature end of the third heat exchanger **42c** and the heat conduction plate **52**. For example, in a case where the heat conduction member is formed of copper or aluminum as described above, the thermal resistance element **60** may be formed of stainless steel (for example, SUS304).

[0060] As shown in the figure, the refrigerant supply line **44** and the refrigerant collection line **46** are connected to an end plate on the low temperature side of the casing of the third heat exchanger **42c**. As described with reference to FIG. **1**, the refrigerant supply line **44** is connected to the high pressure side flow path of the third heat exchanger **42c**, and the refrigerant collection line **46** is connected to the low pressure side flow path of the third heat exchanger **42c**. In addition, the refrigerant supply line **44** is connected to the JT valve **30**.

[0061] Even in this way, during the cool-down, cooling of the third heat exchanger **42c** can be facilitated by the conductive cooling via the second heat conduction path **48b** from the second pre-cooling stage **27**, and the cool-down time of the JT cryocooler **18** can be shortened.

[0062] On the other hand, during the normal operation of the JT cryocooler **18**, a certain degree of temperature difference may occur between the low temperature end of the third heat exchanger **42c** and the second pre-cooling stage **27**. For example, while the low temperature end of the third heat exchanger **42c** is cooled to approximately 5 K, the second pre-cooling stage **27** can be cooled to approximately 12 K as described above. In this case, the second pre-cooling stage **27** serves as a heat source for the third heat exchanger **42c**, and the heat entering the third heat exchanger **42c** from the second pre-cooling stage **27** via the second heat conduction path **48b** may reduce the refrigeration performance of the JT cryocooler **18**.

[0063] However, in the present embodiment, since the thermal resistance element **60** is provided in the second heat conduction path **48b**, it is possible to limit the heat input to the third heat exchanger **42c** via the second heat conduction path **48b** that may occur during the normal operation of the JT cryocooler **18**. Therefore, it is possible to alleviate or prevent the above-described problem.

[0064] In addition, if necessary, the thermal resistance element **60** may be provided in the first heat conduction path **48a**. The first heat conduction path **48a** may connect the second pre-cooling stage **27** to the second heat exchanger **42b** via the thermal resistance element **60**.

[0065] The present invention has been described above based on the examples. It will be understood by those skilled in the art that the present invention is not limited to the above-described embodiment, various design changes are possible, various modification examples are possible, and such modification examples are also within the scope of the present invention. Various characteristics described in relation to one embodiment are also applicable to other embodiments. A new embodiment generated through combination also has the effects of each of the combined embodiments.

[0066] The pre-cooling cryocooler **20** is not limited to the GM cryocooler. The pre-cooling cryocooler **20** may be a cryocooler of another type such as a pulse tube cryocooler or a Stirling cryocooler.

[0067] In the above-described embodiment, a case where the heat exchanger group **42** includes the first to third heat exchangers has been described as an example, but the heat exchanger group **42** may have other multi-stage configurations. For example, the JT cryocooler **18** may include an additional heat exchanger (that is, a fourth heat exchanger) between the third heat exchanger **42c** and the final heat exchanger **32**.

[0068] The JT valve **30** described above may be disposed between the last heat exchanger of the heat exchanger group **42** (that is, the fourth heat exchanger) and the final heat exchanger **32** in the refrigerant supply line **44**. Alternatively, the JT cryocooler **18** may adopt a two-stage JT expansion system, in which a first JT valve is disposed between the third heat exchanger **42c** and the fourth heat exchanger in the refrigerant supply line **44**, and a second JT valve is disposed between the fourth heat exchanger and the final heat exchanger **32** in the refrigerant supply line **44**.

[0069] The JT cryocooler **18** may include a third heat conduction path that connects the second pre-cooling stage **27** to the fourth heat exchanger to enable conductive cooling of the fourth heat exchanger through the second pre-cooling stage **27**. The third heat conduction path may also be the stage extension component **50** and the heat conduction plate **52** as with the first heat conduction path **48a** and the second heat conduction path **48b**. That is, the fourth heat exchanger may be thermally coupled to the second pre-cooling stage **27** via the heat conduction plate **52** and the stage extension component **50**.

[0070] Although the present invention has been described using specific words and phrases based on the embodiment, the embodiment merely shows one aspect of the principle and application of the present invention, and various modifications and improvements can be made within the scope of the present invention described in claims.

[0071] It should be understood that the invention is not limited to the above-described embodiment, but may be modified into various forms on the basis of the spirit of the invention. Additionally, the modifications are included in the scope of the invention.

Claims

1. A Joule-Thomson cryocooler comprising: a pre-cooling cryocooler that comprises a pre-cooling stage; a refrigerant circuit that comprises a heat exchanger and a refrigerant pipe extending from the heat exchanger and cooled by the pre-cooling stage; and a heat conduction path that is provided separately from the refrigerant pipe and connects the pre-cooling stage to the heat exchanger to enable conductive cooling of the heat exchanger through the pre-cooling stage.
2. The Joule-Thomson cryocooler according to claim 1, wherein the pre-cooling cryocooler comprises a cylinder extending from the pre-cooling stage, and the heat conduction path comprises: a first heat conduction member thermally coupled to the pre-cooling stage and extending from the pre-cooling stage to a side opposite to the cylinder, and a second heat conduction member that thermally couples the first heat conduction member to the heat exchanger.
3. The Joule-Thomson cryocooler according to claim 1, wherein the pre-cooling cryocooler comprises a first pre-cooling stage and a second pre-cooling stage that is cooled to a lower temperature than the first pre-cooling stage, the refrigerant circuit comprises a first heat exchanger, a second heat exchanger that further cools a refrigerant cooled by the first heat exchanger, and a third heat exchanger that further cools the refrigerant cooled by the second heat exchanger, and the refrigerant pipe extends from the second heat exchanger via the second pre-cooling stage to the third heat exchanger, and the heat conduction path connects the second pre-cooling stage to at least one of the second heat exchanger and the third heat exchanger to enable conductive cooling of the at least one of the second heat exchanger and the third heat exchanger through the second pre-

cooling stage.

- 4.** The Joule-Thomson cryocooler according to claim 3, wherein the heat conduction path connects the second pre-cooling stage to the second heat exchanger to enable conductive cooling of the second heat exchanger through the second pre-cooling stage.
 - 5.** The Joule-Thomson cryocooler according to claim 3, wherein the pre-cooling cryocooler comprises a cylinder that connects the second pre-cooling stage to the first pre-cooling stage, and the heat conduction path comprises: a stage extension component thermally coupled to the pre-cooling stage and extending from the pre-cooling stage to a side opposite to the cylinder, and a heat conduction plate that thermally couples the stage extension component to the at least one of the second heat exchanger and the third heat exchanger.
 - 6.** The Joule-Thomson cryocooler according to claim 5, wherein the at least one of the second heat exchanger and the third heat exchanger and the stage extension component are disposed on the same side of the heat conduction plate.
 - 7.** The Joule-Thomson cryocooler according to claim 1, wherein the heat conduction path comprises a thermal resistance element.
 - 8.** The Joule-Thomson cryocooler according to claim 2, wherein the heat conduction path comprises a thermal resistance element between the heat exchanger and the second heat conduction member, and a thermal conductivity of the thermal resistance element is smaller than a thermal conductivity of the second heat conduction member.
 - 9.** The Joule-Thomson cryocooler according to claim 3, wherein the heat conduction path connects the second pre-cooling stage to the third heat exchanger via a thermal resistance element.
 - 10.** The Joule-Thomson cryocooler according to claim 5, wherein the heat conduction path comprises a thermal resistance element between the third heat exchanger and the heat conduction plate, and a thermal conductivity of the thermal resistance element is smaller than a thermal conductivity of the heat conduction plate.
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